Week 9 Project: Data-Driven Decision Making: A Cross-Audience Analysis of State Workforce Conditions in the United States

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Purpose and Audiences

The purpose of this analysis is to translate publicly available state-level data on educational attainment, employment, and population into actionable insights for two stakeholder groups with fundamentally different decision-making needs. The underlying data set, drawn from the U.S. Department of Agriculture Economic Research Service, contains educational attainment estimates for 2019 through 2023, labor force and median household income data for the same period, and population estimates from 2020 through 2023. By design, the same integrated data set is used for both audiences; only the framing and presentation change.

The first audience is corporate executives at a multi-state company evaluating where to expand. These decision-makers operate at a strategic altitude. Their time is constrained, their attention is divided across many competing priorities, and they need to make a small number of high-stakes choices about where to allocate capital. They tend to be visually literate but not technically deep, and they want clear rankings, scannable comparisons, and headline takeaways rather than methodological details. Their guiding question is: of all the states the company could expand into, which offer the strongest combination of skilled workforce, healthy labor market, and growing consumer base?

The second audience is local county commissioners. These are elected officials responsible for the long-term economic health of a specific geography. They tend to be more analytically engaged than corporate executives because their decisions have direct consequences for the residents they serve. They need detailed, multi-dimensional context to make policy decisions about workforce development, infrastructure investment, and economic incentives. Their guiding question is different: how does my state compare to the rest of the country across the dimensions that matter most for the people who live here, and where are our most pressing gaps?

These two audiences require fundamentally different visualizations of the same underlying data. The corporate executive wants a clear best answer with minimal cognitive load. The county commissioner wants a comprehensive map of the landscape that supports nuanced interpretation.

Data-Driven Narrative

The data tells a story about uneven workforce development across the United States that has accelerated since 2020. Three publicly available state-level data sets from the U.S. Department of Agriculture Economic Research Service were used as the foundation: educational attainment for 2019 through 2023, labor force and income data for 2019 through 2023, and population estimates for 2020 through 2023. The three files, each originally in long format, were standardized, pivoted to wide format, and merged on the shared FIPS identifier to produce a single state-level analytical table containing roughly 88 attributes for approximately 54 geographic areas.

To provide a unified signal for leadership, I developed a composite Expansion Attractiveness Score (scaled 0 to 1) that blends three key metrics: bachelor's degree attainment, low unemployment, and population growth. This single index eliminates noise by merging workforce talent, labor-market health, and population momentum into one actionable metric.

The story the data tells is one of regional divergence. The states that score highest on the composite measure (Utah, Idaho, South Dakota, Vermont, and New Hampshire) are not the largest economic centers. They are smaller states with educated workforces, low unemployment, and steady inbound migration. Traditional economic giants like California, New York, and Illinois appear in the middle or lower portions of the ranking, weighed down by higher unemployment and net population decline in 2020. Florida is a notable exception, appearing near the top as a large, growing state with low unemployment.

Layering median household income onto the rankings reveals a critical nuance: Maryland has the highest household income in the dataset but ranks only eighth on the composite score. This distinction shifts the strategic conversation. While an executive prioritizing immediate consumer purchasing power would choose Maryland, one focused on long-term growth trajectory would favor Utah or Idaho. The data doesn't make the choice, but it clarifies the trade-offs.

The detailed view of the same data, presented as a four-dimensional bubble scatter for the commissioner audience, surfaces a clear pattern: states with higher educational attainment tend to have lower unemployment rates. The relationship is approximately diagonal. States in the bottom-right quadrant, with high education and low unemployment, represent the strongest workforce conditions. Florida and Texas appear as large green bubbles in or near this zone, signaling population growth despite moderate education levels. New York and California appear as red bubbles in the upper-middle region, reflecting strong education but rising unemployment and population decline. Together, the two visualizations move from a single composite headline to a richly textured map of state-level workforce conditions, and from a strategic question for an executive to a diagnostic resource for a county commissioner.

Evaluation of Visualization Effectiveness

The executive bar chart is highly effective for its intended purpose. By limiting the display to the top 15 states and ordering them by composite score, the chart eliminates the cognitive cost of comparison across all 50 states. The numeric score at the end of each bar provides a precise value for those who want it, while the gradient color encoding allows fast pattern recognition for those who do not. The choice of a sequential color palette (light to dark for low to high median household income) maps cleanly onto the intuition that darker means more, which reduces cognitive load. The visualization has limitations: it does not show which states ranked just below 15th, and it does not reveal the underlying components of the composite score. For an audience that requires a brief overview, however, these omissions are features rather than weaknesses.

The commissioner bubble scatter is effective for a different reason. It encodes four data attributes simultaneously, with education on the x-axis, unemployment on the y-axis, population through bubble size, and population growth through bubble color, and supplements them with median reference lines and quadrant labels. The diagonal pattern that emerges from this view is invisible in the bar chart but central to the story for an analytically engaged audience. Some state labels overlap in the cluster near the medians, and the District of Columbia's outlier position at sixty-five percent bachelor's degrees stretches the x-axis in a way that compresses the central cluster, but both limitations are acceptable trade-offs for the depth of information conveyed. The bar chart integrates two attributes; the scatter integrates four. Both satisfy the multi-attribute requirement, but at different depths consistent with their intended audiences.

Audience Customization and Justification

The two visualizations were intentionally customized for their respective audiences, and each design choice can be justified in terms of how it improves clarity for the target user.

For the executive bar chart, three customization choices stand out. First, the data was filtered to the top 15 states rather than displayed in full, a deliberate reduction in information density because executives evaluating expansion locations rarely consider more than a short list of candidates. Showing all areas would have introduced noise from territories and the national average. Second, the bars were sorted by composite score in descending order with the leader at the top, mirroring how ranked lists are typically read. Third, a second attribute (median household income) was encoded through color rather than as a separate panel, ensuring the income dimension is always anchored to the ranking without forcing the executive to look back and forth between charts.

For the commissioner bubble scatter, the customization choices were oriented toward depth rather than simplicity. Median reference lines were added to give every state a clear positional reference point, so an experienced policy-maker can immediately identify whether their state is above or below the national median on either dimension. Quadrant labels were added in light italic to provide an interpretive frame without distracting from the data itself. The Target Zone label in the bottom-right quadrant uses bold italic and a star symbol to draw the eye toward the configuration that represents the strongest workforce conditions. Bubble size was scaled to make the largest states (California, Texas, and Florida) clearly visible without overwhelming the smaller states. Finally, the diverging red-to-green color palette for population change communicates direction intuitively, with green indicating growth and red indicating decline, and adds a forward-looking dimension to an otherwise static snapshot.

Both visualizations were tuned to their audiences by choosing what to encode, what to suppress, and how to frame the result. The same data drives both charts; only the storytelling layer differs.

Conclusion

This analysis underscores how a single dataset can drive vastly different decisions when tailored to the specific audience. Corporate executives and county commissioners look at the same numbers, but they ask different questions—and require different visual tools. While the executive bar chart compresses a complex composite into a single, ranked baseline for strategic growth, the commissioner bubble scatter expands that same data into a multi-dimensional diagnostic map.

The unifying truth across both perspectives is the sharp regional divergence in U.S. workforce conditions since 2020. Smaller Mountain-West and New England states have pulled ahead by pairing education and low unemployment with strong inbound migration, while several historically dominant state economies have lost ground. For executives, this divergence reveals new market opportunities; for local commissioners, it signals an urgent call to action. In either scenario, the data does not dictate the final decision—it sharpens the strategic question.

AI Usage Acknowledgment

This document and the underlying data pipeline were developed with the assistance of Claude (Anthropic), an AI assistant, in accordance with the course's AI usage guidelines. The AI was utilized as a technical and writing aid to help develop and verify the Python code within the Jupyter notebook, calculate the composite Expansion Attractiveness Score, and review the final manuscript for grammar and clarity. All final analytical, programming, and editorial decisions are entirely my own.

References

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